ALAHMEDS STUDY / MACROECONOMICS

# Midterms 1-15 | Formulas and graph rules

One-page landscape study reference. Coverage: 15 of 28 modules. Generated 11 Nov 2026.

## Prices and real output [M1]

$\pi = (P_{new}/P_{old} - 1) \times 100\%$

P price index; π period inflation.

Nominal GDP = $\Sigma\, p_t q_t$

Real GDP = $\Sigma\, p_{base} q_t$

p price, q quantity, Σ sum. Hold base prices fixed.

Fixed-basket index = $(cost_t/cost_{base}) \times 100$

Same basket. Index 106 means price ratio 1.06.

GDP deflator = (nominal GDP/real GDP) × 100

Real GDP = nominal GDP / (deflator/100).

$g_{real} = (1+g_{nominal})/(1+\pi_{GDP}) - 1$

Decimals. Approx.: real = nominal growth minus inflation.

SRC-011 slides 52-57; SRC-079 pp.1-5; SRC-129 p.2.

## Production and growth [M4-5]

$Y = A \times K^{0.3} \times N^{0.7}$

Y output/period; K capital stock; N hours/period; A efficiency.

$ALP = Y/N = A(K/N)^{0.3}$

ALP = output/hour. Fixed A,K: N ↑ → Y ↑, ALP ↓.

$Y'/Y = (A'/A)(K'/K)^{0.3}(N'/N)^{0.7}$

Exact growth factors; prime = new. Fixed A: scale K,N together → same scale for Y.

$g_Y \approx g_A + 0.3g_K + 0.7g_N$

g = growth rate. Small-change approximation; match periods and rate units. Use exact factors for large changes.

$g_A \approx g_Y - 0.3g_K - 0.7g_N$

TFP growth is the residual; A ≠ output/hour. Higher K/N raises ALP at fixed A.

$g_{Y/N} \approx g_A + 0.3(g_K - g_N) \approx g_Y - g_N$

Approximate output/hour growth. g = growth rate; match periods and rate units.

SRC-012 slides 4-8,12-15; SRC-082; SRC-129 pp.3-7. Ratios derived.

## Potential output and graph axes [M11]

$Y^* = A\,K^{0.3}(N^*)^{0.7}$

N* clearing hours; Y* not a physical maximum.

Labor: W/P vertical; N horizontal

Demand down; supply up, lifetime resources fixed.

LRAS: P vertical; Y horizontal; Y = Y*

Flexible W: double P,W → same real wage, N*,Y*.

A,K or clearing N* changes → LRAS can shift

P alone: along LRAS. Sticky W may put Y below Y*.

SRC-012 slides 4,8,19-29,52-59; SRC-129 p.13.

## GDP and saving [M2,5]

Y = C + I + G + NX; NX = EX − IM

C consumption; I physical investment; G purchases; EX exports; IM imports.

I: new fixed capital + homes + Δinventories

Δ change. Transfers are not G; asset trades are not new output.

Value added = output − intermediate inputs

GDP = value added = final output = spending = income. Count once.

Imported phone: C +500, NX −500 ⇒ ΔY = 0

Domestic services count separately. Identity ≠ causal prediction.

S = Y − C − G = I + NX; NX = S − I

S = national saving. All quantities are flows per period.

SRC-011 slides 34-42,47-50; SRC-081 pp.1-3; SRC-140 Q1; SRC-012 slide 13.

## Marginal products [M4,6]

$MPN = 0.7\,A(K/N)^{0.3} = 0.7Y/N$

Marginal output/hour; local gain.

$MPK = 0.3\,A(N/K)^{0.7} = 0.3Y/K$

Own input ↑ lowers own MP. K ↑ raises MPN; N ↑ raises MPK.

Competitive firm: MPN = w = W/P

w goods/hour. MPN > w: hire; MPN < w: reduce hours.

Labor income = 0.7Y; capital income = 0.3Y

Competitive payments. A or K ↑: labor demand right.

SRC-012 slides 4-7,18-20; SRC-082; SRC-129 pp.3-7.

## Household budget [M13]

$c_f = (a+y-c)(1+r)+y_f$

a net assets (debt negative); y disposable real income; c consumption; f future. All real.

$PVLR = a+y+y_f/(1+r)$

$PVLC = c+c_f/(1+r) = PVLR$

PVLR resources; PVLC spending, valued today. Match rate and period.

$MU(c) = \beta(1+r)MU(c_f)$

MU marginal utility; β future weight. Same utility; β(1+r)=1 gives c=c_f.

$c = c_f = PVLR/[1+1/(1+r)]$

Equal-consumption case; free borrowing/lending and β(1+r)=1.

$s = y-c$; $a_{next} = (a+s)(1+r)$

s saving flow; a_next net assets after interest. Count initial a once.

SRC-143 slides17-22; SRC-086; SRC-129 pp.16-25.

## Employment denominators [M3]

LF = E + U; pop = E + U + outside

E employed; U jobless, searching. pop = eligible civilian non-institutionalized people age 16+.

Unemployment = U/LF × 100%

Denominator LF. Search exit lowers U and LF; lower rate need not mean more jobs.

Participation = LF/pop × 100%

Denominator pop (people), not price level P. LF includes E and U.

Employment/population = E/pop × 100%

Household: people; payroll: covered jobs. Zero cyclical can leave frictional/structural unemployment.

SRC-011 slides 25-31. pop is teaching shorthand.

## Work reward and total hours [M7,10]

$w = W/P$; $R = [(1-t_n)/(1+t_c)]w$

W $/hour; P $/good; w,R goods/hour. t_n labor, t_c sales tax; decimals.

w=10: labor tax 25% → 7.5; sales tax 25% → 8

Separate: 10×.75 versus 10/1.25; P excludes sales tax.

$N_s(w) = \Sigma_j h_j(w)$; equal h: $N_s = pop \times h$

Sum hours at the same w; h hours/person.

$N_d(w^*) = N_s(w^*) = N^*$

* equilibrium; N total hours, not people.

SRC-012 slides 21-27,32-35; SRC-083 pp.2-8,11. Tax ratio/sum derived.

## Income news and time [M14-15]

c = (a + Σ remaining y)/L

r=0, β=1; free borrowing; equal periods; zero final assets. L periods left.

MPC = Δc(today)/Δy(today)

Dollar changes, stated experiment. Δ means change, not percent.

One-time Δy today: Δc = Δy/L; MPC=1/L

Every remaining period Δy: Δc=Δy; MPC=1

Surprises, assumptions above. One-time asset gain: Δc=Δa/L.

Replan: $c_{new} = [a_{now} + \Sigma \text{ remaining expected } y_{new}]/L_{left}$

Inherited debt: negative a_now. Do not reuse past resources or expected news.

Keynesian $c = a_0 + b\,y$; MPC=b

a0 autonomous spending; b current-income MPC. Distinct model.

SRC-143 slides15,22-26; SRC-133; SRC-127 extensions. Remaining-period formula: derived.

## Buying power and interest [M1]

Goods bought = dollars / (dollars per good)

Fixed dollars: a higher price buys fewer goods.

$r^e \approx i - \pi^e$; $r^a \approx i - \pi^a$

i = nominal rate; e expected; a actual. Match periods.

$1+r^a = (1+i)/(1+\pi^a)$

Exact; decimal rates. Surprise inflation benefits fixed-debt borrowers, hurts lenders.

SRC-011 slides 59-62; SRC-129 p.2. Exact ratio: derived.

## Capital and investment [M12]

$P \times MPK_{next} = p(k)(r+\delta)$

If P=p(k): $MPK_{next} = r+\delta$

Expected MPK. P output price; p(k) capital price. Same-period real r,δ; no spread.

$K' = (1-\delta)K + I$; $I = K' - (1-\delta)K$

K current, K′ next stock; I gross flow; δ share (=d).

Net $I = I - \delta K = K' - K$

δK replacement; gross I > 0 may shrink K.

r ↓ → desired K′ ↑ → I ↑ (current K fixed)

r vertical, I horizontal. Expected A ↑ shifts demand right; spread ↑ raises cost.

SRC-143 slides 4-13; SRC-085 pp.1-7; SRC-129 p.15.

## Labor changes: horizon matters [M7-10]

w ↑ → $N_s$ ↑, PVLR fixed; PVLR ↑ → $N_s$ ↓, w fixed

PVLR lifetime resources; normal leisure. Movement versus wealth shift.

Temporary A ↑: $N_d$ →; N*, w*, Y ↑

K fixed; negligible wealth effect. Leisure value ↑: supply left; N,Y ↓; w ↑.

Permanent A ↑: demand →; wealth effect on supply ←

Normal leisure: N ambiguous, w ↑; course Y ↑.

Temporary tax ↑: $N_s$ ←; N* ↓; w* ↑; R ↓

A,K fixed; negligible wealth effect; no offsetting transfer.

Permanent tax ↑: substitution ←; income →

Normal leisure, no transfer: net N may ↓, stay or ↑.

Population ↓ → aggregate $N_s$ ←

Remaining individuals can keep the same supply schedule.

Value of leisure ↑ → individual + aggregate supply ←

Demand fixed: N* ↓, w* ↑. Wage alone moves along supply.

SRC-012 slides 21-35,39,54; SRC-083 pp.2-8,11; SRC-131.

Exam: one handwritten standard-size sheet, front only; no photocopy. This PDF is a study/hand-copy guide. [SRC-127]